National University of Computer and Emerging Sciences
Lahore Campus

µP Interfacing and Programming (EE3002/EE3012)

Date: 19th September, 2026

Course Instructor(s)

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Sessional-I Exam

Total Time (Hrs):	**1**
Total Marks:	**30**
Total Questions:	**2**

[redacted]
Roll No Section

Student Signature

Do not write below this line

Instructions/Notes:
1. **Attach question paper with answer sheet.**
2. **Do not forget to write your Name and Roll Numbers.**
3. **Make assumptions if you think some data is missing.**
4. **Solve on the answer sheets. Answers written on the question paper will not be marked.**

CLO 1: **Describe** the internal architecture, programming of microprocessor and microcontroller. **(PLO 1, C2)**

Q1: [5+5+5 marks]

a) **Explain** the difference between von Neumann and Harvard architectures. **Illustrate** your answer with neat and labeled diagrams, and **describe** the advantages and disadvantages of each architecture.

b) The PIC16F84 microcontroller has a File Register (data memory) space of 80 bytes, consisting of 12 bytes allocated to Special Function Registers (SFR) and the remaining bytes allocated to General Purpose Registers (GPR). **Determine** the required size (number of bits) of the address bus needed to access this memory, and **identify** the address range (locations) assigned to the SFR and the GPR.

c) **Differentiate** between RISC and CISC architectures. Suggest practical methods that microprocessor manufacturers can adopt to enhance CPU processing power.

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CLO 1: **Describe** the internal architecture, programming of microprocessor and microcontroller. **(PLO 1, C2)**

Q2: **[8+7 marks]**

a) A traffic signal controller uses a PIC18 microcontroller with an 8 MHz crystal oscillator. **Explain** the procedure and write a delay subroutine in Assembly Language that produces an approximately 2-seconds delay. **Show** all calculations.

b) **Explain** the operation of the PIC18 hardware stack during the execution of the subroutines in the given program. Draw the hardware stack, **depicting** its contents, and indicate the value of the Stack Pointer at all the appropriate stages of subroutine execution.

```
COUNT EQU 0x07
MYREG EQU 0x08
MYREG2 EQU 0x09
000000              ORG 0
000000              MOVLW 0
000002              MOVWF COUNT
000004    BACK      CALL DISPLAY
000008              GOTO BACK
00000C    DISPLAY   INCF COUNT, F
00000E              MOVFF COUNT, PORTB
000012              CALL DELAY1
000016              CALL DELAY2
00001A              RETURN
;--------- first delay subroutine
000300              ORG 300H
000300    DELAY1    MOVLW 0xFF
000302              MOVWF MYREG
000304    AGAIN1    NOP
000306              NOP
000308              NOP
00030A              DECF MYREG, F
00030C              BNZ AGAIN1
00030E              RETURN
; --------- second delay subroutine
000400              ORG 400H
000400    DELAY2    MOVLW 0x80
000402              MOVWF MYREG2
000404    AGAIN2    NOP
000406              NOP
000408              DECF MYREG2, F
00040A              BNZ AGAIN2
00040C              RETURN
END
```


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